Manchester Metropolitan University

Faculty of Science and Engineering
Department of Computing and Mathematics
Level 4

Calculus : Mock exam 02

Instructions to candidates:

Answer **ALL** questions in **SECTION A**.

Answer **TWO** questions in **SECTION B**. If you answer more than two questions from section B your mark will be calculated using the best two questions.

Permitted materials:

- Calculators are permitted.

SECTION A

Answer **ALL** questions in this section.

1. (a) Give the ϵ, N definition for the convergence of an infinite sequence, $\{x_n\}_{n=1}^{\infty}$, to a limit, L. Comment briefly on the roles played by the parameters ϵ and N in this definition. (5 marks)

   (b) State the definition, in terms of partial sums, of the convergence of an infinite series, $\sum_{n=1}^{\infty} x_n$. Then explain briefly why the *General Term Test for Divergence* is true, that if the sequence of terms $\{x_n\}_{n=1}^{\infty}$ does not converge to zero, the series must diverge. (5 marks)

   (c) Consider the sequence $\{x_n\}_{n=1}^{\infty}$ defined by $x_n = \frac{1}{2n-1}$. Using the ϵ, N definition of convergence, prove that $\lim_{n\to\infty} x_n = 0$. (8 marks)

   (d) Consider the series $\sum_{n=1}^{\infty} a_n$, where the term is defined by $a_n = \frac{3n^2+1}{2n^2}$. Use the general term test to prove that this series diverges.

   (7 marks)

SECTION A

2. (a) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous function. Using the concept of limiting value, give the formal definition of the derivative, $f'(a)$, of f evaluated at the point a.

    Comment briefly on the geometric interpretation of the derivative, how it relates to the graph of the function f. (5 marks)

   (b) Give the definition of an anti-derivative, F, of a function, $f : \mathbb{R} \to \mathbb{R}$. How will two different anti-derivatives of f be related to each other? Justify your answer.

    (5 marks)

   (c) Consider the function $g : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ defined by $g(x) = \frac{1}{x}$. Using the definition of the derivative you gave in part (a), prove that $g'(x) = -\frac{1}{x^2}$.

    (7 marks)

   (d) Use the principle of anti-differentiation to find the unique function $h : \mathbb{R} \to \mathbb{R}$ that satisfies $h''(x) = -\sin(x)$ and has the initial values $h'(0) = 1$ and $h(0) = 0$. (8 marks)

END OF SECTION A

SECTION B

Answer **TWO** questions in this section.

3. (a) Let the sequence $\{c_m\}_{m=1}^{\infty}$ be defined by the formula

$$c_m = \frac{3m^4 + 2m^2 + 1}{-7m^4 + 8}.$$

Show how the algebra of limits theorem for convergent sequences can be used to prove that $\{c_m\}_{m=1}^{\infty}$ is a convergent sequence and to determine its limit. Make it clear in your answer where you are using each part of the theorem that you rely on. (7 marks)

(b) Use the ϵ, N definition of sequence convergence to prove the product part of the algebra of limits theorem for convergent sequences. That is, prove that if $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are convergent sequences with limits L and M respectively (with L and M assumed to be non-zero), then the *product* sequence $\{x_n y_n\}_{n=1}^{\infty}$ is a convergent sequence with limit LM. (7 marks)

(c) Consider the sequence $\{a_n\}_{n=1}^{\infty}$ defined by $a_0 = 1$ and the recurrence relation

$$a_n = \frac{1}{5 - a_{n-1}}, \quad \text{for } n \geq 1.$$

In addition, assume that

- the sequence $\{a_n\}_{n=1}^{\infty}$ is bounded below by 0, and
- the sequence $\{a_n\}_{n=1}^{\infty}$ is monotone decreasing.

Use the Monotone Convergence Theorem to deduce that the sequence $\{a_n\}_{n=1}^{\infty}$ is a convergent sequence and then find its limit. Explain your reasoning carefully and point out any properties of sequence convergence that you use. (7 marks)

(d) Prove the *monotone decreasing* assumption from part (c). (4 marks)

SECTION B

4. (a) Consider the infinite series

$$\sum_{n=1}^{\infty} \frac{-2+3^n}{5^{n+3}}.$$

Use the result about the convergence of geometric series to prove that the series defined above is convergent and determine its sum. You can use the theorem about linear combinations of convergent series. (8 marks)

(b) Use a suitable comparison test to prove that the series

$$\sum_{n=1}^{\infty} \frac{3n^2}{9-n^4},$$

is a convergent series. You can make use of the known convergence of any hyper-harmonic series. (5 marks)

(c) Use the ratio test to prove that the series

$$\sum_{m=1}^{\infty} \frac{5^m}{(m+3)!},$$

is a convergent series. *Note that* ! *denotes the factorial operation*.

(5 marks)

(d) Obtain the partial fraction expansion of

$$\frac{21}{m^2+3m},$$

and use it to obtain the sum of the infinite series

$$\sum_{m=1}^{\infty} \frac{21}{m^2+3m},$$

by considering cancellation effects in the partial sums of this series.

(7 marks)

SECTION B

5. (a) Use the limit definition of the derivative to prove the product rule for differentiation. That is, prove that if f and g are differentiable functions, then the derivative of their product is given by

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Point out clearly where you make use of any properties of the convergence of functions. (5 marks)

(b) Consider the function g defined by the formula

$$g(x) = 2x^4 \cos(3x^2).$$

Use the linearity, product and chain rules to obtain the first and second order derivatives, $g'(x)$ and $g''(x)$, of g. Point out clearly where you are using each of the rules. (5 marks)

(c) Consider the graph in the xy-plane defined by the equation $y = \sqrt{x}$, for $x \geq 0$. Use the concepts of *differentiation* and *stationary point* to find the point on this curve that is closest to the point $(5, 0)$. (10 marks)

(d) Use a combination of the product rule and quotient rule to fully expand the derivative F', of a function F defined by

$$F(x) = \frac{u(x)}{v(x)w(x)z(x)},$$

and show how it depends on the individual functions u, v, w, z and their derivatives. You can assume that v, w and z are always nonzero.

(5 marks)

SECTION B

6. (a) Show how integration by parts can be applied to evaluate the definite integral

$$I_1 = \int_{-3\pi/2}^{3\pi/2} e^{2x} x^2 \, dx.$$

(6 marks)

(b) Use the substitution $u = x + 5$ to evaluate the indefinite integral

$$I_2 = \int \frac{2x^2 + 3x + 1}{x + 5} \, dx.$$

(6 marks)

(c) The Riemann integral of a function $f(x)$ between the limits $x = a$ and $x = b$ is defined as

$$\int_a^b f(x) \, dx = \lim_{\substack{n \to \infty \\ \Delta x \to 0}} \left(\sum_{i=1}^{n} f(x_i^*) \Delta x_{i-1}, \right).$$

Set up and evaluate the limit in the Riemann integral to show that

$$\int_0^5 x^3 \, dx = \frac{625}{4}.$$

In your answer you should clearly define the partition of the interval $(0, 2)$ you are using and what the objects $x_i^*, \Delta x_{i-1}$ and Δx are, and point out any properties of the limit process you use.

You can use, without proof, the summation formula

$$\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4}.$$

(13 marks)

END OF SECTION B
END OF QUESTIONS